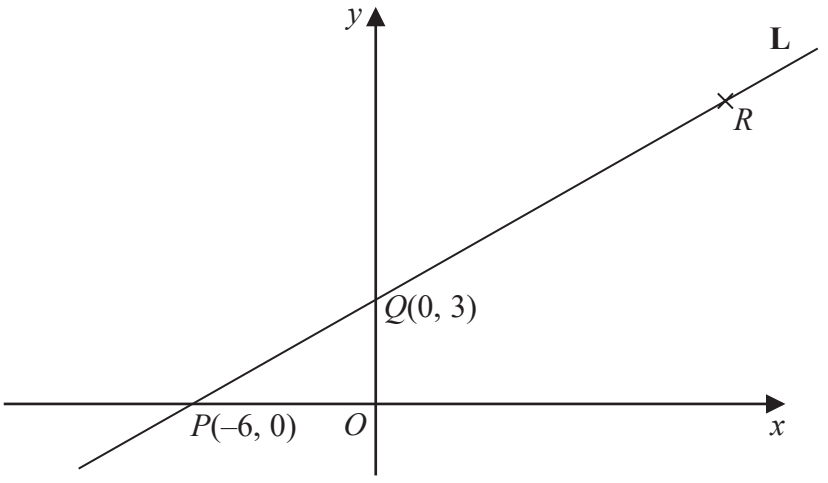
11 Here is a sketch of the line **L**.



The points $P(-6, 0)$ and $Q(0, 3)$ are points on the line **L**.

The point R is such that PQR is a straight line and $PQ : QR = 2 : 3$

(a) Find the coordinates of R.

P to Q: (6, 3). QR = (3/2)(6, 3) = (9, 4.5).

R = Q + QR = (0, 3) + (9, 4.5) = (9, 7.5).

(9, 7.5)

1 mark
QR = 3/2 of PQ

1 mark
R=(9,7.5)

(2)

(b) Find an equation of the line that is perpendicular to **L** and passes through Q.

1 mark
gradient of L = 1/2

1 mark
perpendicular = -2

1 mark
y=-2x+3

Gradient of L = (3 - 0)/(0 - (-6)) = 1/2.

Perpendicular gradient = -2.

Through Q(0,3): y = -2x + 3.

y = -2x + 3

(3)

(Total for Question 11 is 5 marks)

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